

IoT FALL DETECTION SYSTEM

Using MPU6050, ESP32, And Blynk—Safeguard Your Loved Ones!

YOGESH BAWANE

Ideal for those seeking to enhance safety for loved ones or individuals at risk of falls, this IoT fall detection system provides a proactive solution. Explore the realm of IoT and its potential impact on our lives.

Falls pose a serious risk, particularly for the elderly and those with medical conditions. With this IoT-based fall detection system, instant notifications on a mobile phone can be received in the event of a fall. This enables a swift response, potentially preventing serious repercussions.

Utilising a six-axis gyro cum acceleration sensor and a microcontroller, we can detect and measure motion and orientation changes, and with Blynk, a user-friendly interface can be created for receiving notifications and alerts. The hardware and software implementation, including setting up the MPU6050 sensor and ESP32 microcontroller, configuring the Blynk app, and programming the system using the Arduino IDE, is explained in detail. Fig. 1 displays the author's prototype. Components used in the system are listed in the bill of materials table.

BILL OF MATERIALS

Components	Description
ESP32 (MOD1)	Microcontroller for programming
MPU6050 (MOD2)	6-axis gyro and acc. sensor
Micro USB	
Breadboard (for circuit prototyping)	
Male-to-male jumper wires	

Circuit diagram and working

The circuit diagram of the IoT fall detection system using MPU6050,

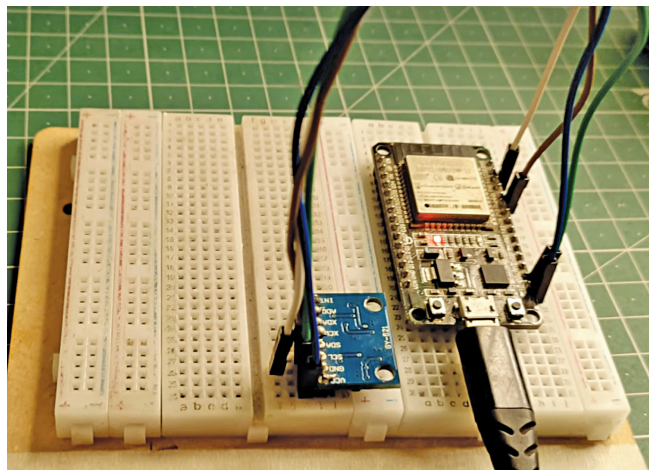


Fig. 1: Author's prototype

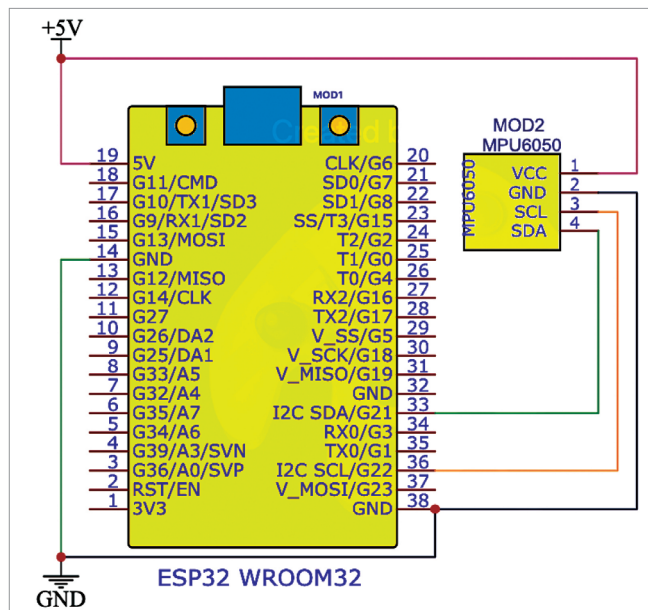


Fig. 2: Circuit diagram

ESP32, and Blynk is shown in Fig. 2. It comprises ESP32 microcontroller (MOD1), MPU6050 (MOD2), micro USB cable, breadboard, and a few male-to-male jumpers.

Before coding, set up the Blynk API key and account by following the instructions in the link <https://www.youtube.com/watch?v=ycAf2J4LL3o>.

First, create the Blynk account, then create the project template, obtain the authentication token, and note it for use in the code later. Also, add the required libraries. Fig. 3 shows the Blynk dashboard, and Fig. 4 shows how the Blynk setup is done.

Software coding

For code, use the following libraries and install them using the library